

UrbanPulse: Satellite Urban Sprawl & Land Cover Intelligence

High-resolution multi-temporal satellite analytics engine tracking urban sprawl, concentric ring densification, and Shannon entropy across metropolitan regions using Sentinel-2 and ESA WorldCover.

[🌐 Live Web Dashboard](#)[💻 GitHub Repository](#)[📅 Generated: 2026-10-03 19:08 UTC](#)[🔴 Git Commit: aed529b0](#)[✓ CI/CD: Passing](#)

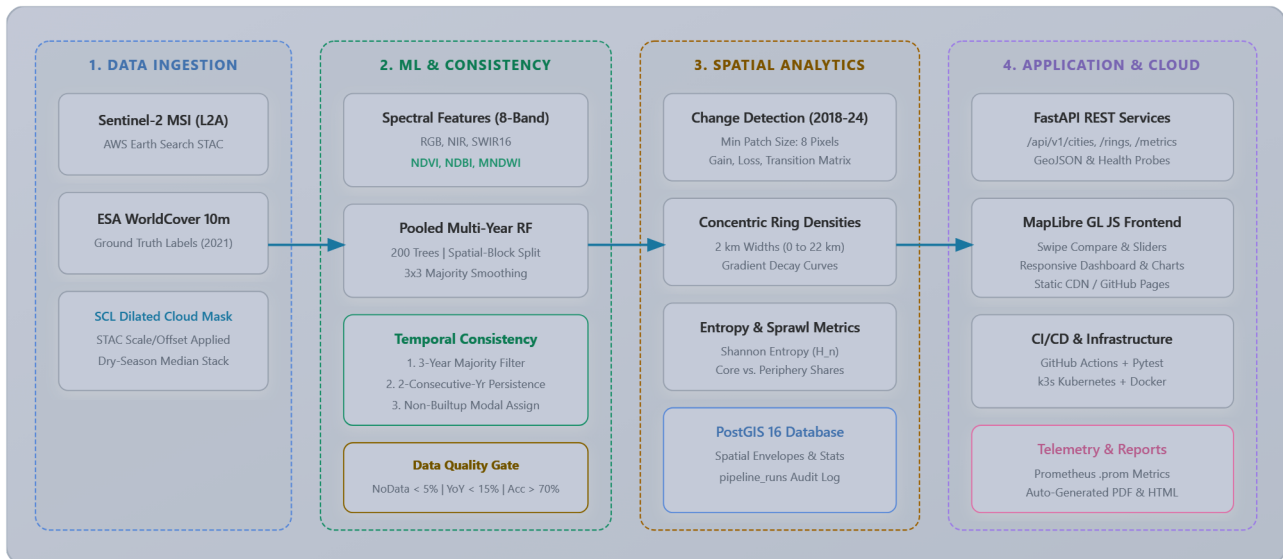
1. Problem & Approach

Rapid urbanization across major metropolitan regions frequently outpaces civic planning frameworks, leading to unchecked peripheral sprawl, fragmentation of agricultural zones, and loss of critical blue-green infrastructure. Monitoring these dynamics using ad-hoc, single-scene satellite imagery often introduces severe seasonal phenological confusion, cloud contamination, and sensor baseline artifacts that corrupt time-series measurements.

UrbanPulse was engineered as an automated, reproducible Earth Observation (EO) pipeline. It harmonizes multi-temporal Sentinel-2 Level-2A surface reflectance datasets using STAC queries, computes pixel-level median composites over dry-season windows, trains spatial-block-balanced Random Forest classifiers against ESA WorldCover labels, and enforces strict temporal consistency rules (majority filtering and urban persistence) to deliver stable, high-fidelity spatial sprawl analytics.

✂ 2. System Architecture

The end-to-end data flow spanning satellite catalog ingestion, radiometric processing, machine learning classification, PostGIS spatial ingestion, and containerized deployment.





3. Data & Methodology

Earth Observation Data Specifications

- **Sensor:** Sentinel-2 Multi-Spectral Instrument (MSI), Level-2A Bottom-Of-Atmosphere (BOA) surface reflectance via AWS Earth Search STAC.
- **Spatial Resolution:** 60.0 m analytical pixel grid (EPSG:32643 UTM projection), with 10.0 m high-resolution visual previews.
- **Area of Interest (AOI):** 45.0 × 45.0 km bounding envelope centered on the municipal core (571,536 grid pixels per city).
- **Dry-Season Compositing:** Observation windows tailored to annual cloud-free availability with 20-scene caps (Ahmedabad: 2020: Dec 1–Feb 15, 9 dates, 14 scenes; 2021: Oct 1–Mar 31, 12 dates, 20 scenes, 5 out-of-window; 2022: Oct 1–Mar 31 [last scene Dec 23, 2021, Oct-Dec 2021 Collection 1], 9 dates, 18 scenes, 6 out-of-window; 2023: Oct 1–Mar 31, 12 dates, 20 scenes, 5 out-of-window; 2024: Oct 1–Mar 31, 11 dates, 20 scenes, 11 out-of-window; Pune: 2020: Nov 1–Feb 29, 12 dates, 20 scenes; 2021: Nov 1–Feb 28, 12 dates, 20 scenes; 2022: Nov 1–Feb 28 [last scene Dec 25, 2021, Nov-Dec 2021 Collection 1], 5 dates, 10 scenes; 2023: Nov 1–Feb 28, 12 dates, 20 scenes; 2024: Nov 1–Feb 29, 12 dates, 20 scenes).
- **Cloud & Shadow Masking:** SCL (Scene Classification Layer) masking cloud shadow, medium/high probability cloud, cirrus, and snow with 1-pixel binary dilation.
- **Radiometric Calibration:** Scale (0.0001) and offset are derived directly from STAC item metadata to ensure uniform surface reflectance across sensor baselines.

Machine Learning & Temporal Consistency

- **Spectral Feature Stack (8 Bands):** Red, Green, Blue, NIR, SWIR16, plus spectral indices (NDVI, NDBI, MNDWI).
- **Supervised Classifier:** Random Forest (200 trees, `class_weight="balanced"`) pooled across reference years (2018, 2021, 2024).
- **Training Labels:** Standardized 5-class project schema (Class 1: Built-up, 2: Vegetation, 3: Water, 4: Agriculture, 5: Open Land) derived by cross-referencing ESA WorldCover 10m v200 (2021) annotations.
- **Spatial Block Partitioning:** 8-fold non-overlapping spatial grid partition guaranteeing zero spatial autocorrelation between train and test folds.
- **Spatial Filtering:** 3 × 3 majority mode filter applied to raw raster predictions to eliminate salt-and-pepper noise.
- **Temporal Consistency Rules:**
 1. *Majority Rule:* Pixel is Built-up in year t only if Built-up in ≥ 2 of $(t-1, t, t+1)$.
 2. *Urban Persistence:* Once Built-up for 2 consecutive years, pixels permanently stay Built-up (urban land persistence assumption).

3. *Modal Non-Built-Up*: Filtered false-urban pixels receive their multi-year non-built-up modal class.

Methodological Disclosure: Ahmedabad's 2021, 2023 and 2024 composites use the 20 lowest-cloud scenes between October and March, so they include October and March scenes (11 of 20 in 2024). Pune's composites use November-February only. Ahmedabad's 2020 composite was built earlier with a strict Dec 1-Feb 15 window; current code does not reproduce its exact scene list. How much the window affects Ahmedabad's built-up area has not been tested.

4. Multi-City Analytical Results

Metropolitan Region: Ahmedabad

Model Evaluation & Accuracy (Pooled Multi-Year Random Forest)

OVERALL TEST ACCURACY

65.25%

Held-out Spatial Blocks

COHEN'S KAPPA (K)

0.5600

Inter-Class Agreement

PER-YEAR ACCURACIES

2018: 59.5% • 2021: 68.5% • 2024: 67.8%

Per-Class Precision, Recall & F1-Score

Class Name	Precision	Recall	F1-Score	Support
Built-up	72.2%	79.5%	75.6%	258
Vegetation	59.2%	52.2%	55.5%	270
Water	87.4%	89.0%	88.2%	219
Agriculture	52.3%	59.5%	55.6%	291
Open land	55.6%	42.6%	48.3%	162

5 × 5 Confusion Matrix

True \ Pred	Built	Veg	Water	Agri	Open
Built-	205	10	1	18	24
Vegeta	16	141	10	93	10
Water	6	8	195	9	1
Agricu	15	68	15	173	20
Open l	42	11	2	38	69



2020–2024 Headline Urban Expansion

2020–2024 EXPANSION RANGE

+53.0 to +84.0 km²

Across Processing Methods

ESA WORLDCOVER 2021 ANCHOR

393.73 km²

Independent Benchmark (2021)

2024 CLEANED BUILT-UP FOOTPRINT

468.54 km²

Analysis Window: 2020–2024

Framing Note: Analysis is framed over **2020–2024**. Pre-2020 observations (2018–2019) are omitted due to cloud coverage and early baseline variability. The year **2022** is designated *Oct-Dec 2021, Collection 1* (dry-season slice from early Collection 1 archive; scale and offset are derived directly from STAC item metadata).

Multi-Series Growth & Method-Sensitivity Band (2020–2024)

Year	Cleaned Series (km ²)	Raw Classified (km ²)	TLS Normalised (km ²)
2020	384.51 km ²	421.06 km ²	415.97 km ²
2021	414.07 km ²	413.61 km ²	413.61 km ²
2022 (Oct-Dec 2021, Collection 1)	441.91 km ²	423.91 km ²	426.94 km ²
2023	474.54 km ²	504.42 km ²	451.61 km ²
2024	468.54 km ²	474.02 km ²	474.88 km ²

Leave-One-Year-Out (LOYO) Validation (Evaluated on ALL Held-Out Points)

To evaluate temporal stability without data leakage, classifiers were trained excluding each fold year, and tested on **all held-out points** (not restricted to stable points). Area estimates are adjusted using Olofsson et al. (2014) area-weighted stratified estimation with 95% confidence intervals.

Held-Out Year	Processing Stage	Precision	Recall	Built-up F1	Mapping Accuracy
2018	Raw (Before)	68.5%	73.3%	0.7079	
2018	TLS Normalized (After)	74.2%	76.7%	0.7543	
2021	Raw (Before)	80.0%	79.1%	0.7953	
2021	TLS Normalized (After)	77.9%	77.9%	0.7791	
2024	Raw (Before)	68.9%	82.6%	0.7513	
2024	TLS Normalized (After)	68.0%	76.7%	0.7213	

⚠ Negative Result Notice (Radiometric Normalisation):

Cross-year Total Least Squares (TLS) pseudo-invariant feature radiometric normalisation was implemented and systematically tested. Empirical evaluation across all held-out test

points demonstrates that **radiometric normalisation did not reduce year-to-year drift** compared to the baseline composite series. Consequently, rule-based temporal consistency filtering remains the authoritative operational mechanism for ensuring monotonic urban expansion.

4-Stratum Change Validation (2020–2024) & Error-Adjusted Areas

To independently validate land cover transitions between 2020 and 2024, a probability sample of **N=300** verification points across 4 spatial strata (Gain, Persistent Built, Persistent Non-built, Loss) was audited following the design-based paradigm of **Olofsson et al. (2014)**.

Stratum	Mapped Area (km ²)	Evaluated (N)	(0,0) Non-built	(0,1) Gain
A – Stratum A (Mapped Gain)	116.65 km ²	99	40	41
B – Stratum B (Persistent Built)	358.23 km ²	50	2	1
C – Stratum C (Persistent Non-built)	1635.22 km ²	98	94	2
D – Stratum D (Mapped Loss)	57.74 km ²	50	45	0

ADJUSTED GROSS GAIN

88.85 ± 49.48 km²

95% CI: [39.36, 138.33] km²

ADJUSTED GROSS LOSS

7.16 ± 35.28 km²95% CI: [0.00, 42.45] km²

ADJUSTED NET CHANGE

+81.68 ± 51.56 km²95% CI: [+30.12, +133.24] km²**Sensitivity & Scenario Analysis:**

- **Pre-recheck Baseline:** Net Change = +14.90 ± 83.60 km² (dominated by false loss variance in Stratum C).
- **Without Stratum C Gains (IDs 50 & 172 treated as errors):** Net Change = +48.31 ± 23.10 km².
- **Post-recheck Adjusted Net (Primary):** Net Change = +81.68 ± 51.56 km² (95% CI: [+30.12, +133.24] km²).

✓ Verification Summary:

Validated on 297 points (Ahmedabad only); net change is distinguishable from zero post-recheck (+81.7 ± 51.6 km²), but depends on the recheck (pre-recheck: +14.9 ± 83.6 km²; without Stratum C gains: +48.3 ± 23.1 km²).

Labelling Protocol: Single independent human interpreter; paired Sentinel-2 10m dry-season RGB surface reflectance composites (2020 vs 2024) corroborated against high-resolution Google Earth Pro historical timeline imagery. Ambiguous points with mixed-pixel confusion or low visual contrast were marked as *unclear* (3 points) and excluded from primary estimation.

Labelling History: (1) First pass was conducted on initial 2024 chips; (2) Second pass was conducted with corrected reflectance stretch chips (26 of 291 start labels changed: 12 built to not built, 14 not built to built, establishing the labeller's change rate at ~9%, and 30 end labels changed from not built to built); (3) Third pass rechecked 10 discordant points (apparent losses and gains) against Google Earth Pro historical timeline imagery, updating 6 labels (IDs 81, 91, 94, 123 in Stratum C; IDs 151, 226 in Stratum D). The 6 corrected labels persisted through the second pass, and concordant points were not rechecked. These initial discrepancies were labelling errors in the first pass (bare or ploughed soil read as built-up on 2020 Sentinel-2 chips), corrected using Google Earth historical imagery (not classifier errors).

Disclaimer: Intervals reflect sampling error only; labelling inconsistency (about 9% between passes) is not included.

**Methodology Notes:* Strata were generated from annual TLS-normalised classifications (ahmedabad_2020_classified.tif and ahmedabad_2024_classified.tif) with a 3x3 majority filter (no temporal consistency rules). Mapped built-up areas (415.97 km² in 2020 and 474.88 km² in 2024) match the dashboard TLS series (415.97 and 474.88 km²), both produced by the 3x3 majority filtered TLS classification pipeline without temporal filtering. Gross loss standard error applies Laplace (add-one) smoothing for zero-count sample proportions. Validated series is the TLS-normalised classification.

Spatial Sprawl, Entropy & Core-Periphery Distribution

Year	Cleaned Built-up (km ²)	Shannon Entropy (H _n)	Core Share (0-6 km)
2020	421.06 km ²	0.9282	26.3%
2021	413.61 km ²	0.9354	25.3%
2022 (Oct-Dec 2021, Collection 1)	423.91 km ²	0.9423	24.1%
2023	504.42 km ²	0.9477	22.3%
2024	474.02 km ²	0.9469	22.5%

Metropolitan Region: Pune

Model Evaluation & Accuracy (Pooled Multi-Year Random Forest)

OVERALL TEST ACCURACY

81.96%

Held-out Spatial Blocks

COHEN'S KAPPA (K)

0.7582

Inter-Class Agreement

PER-YEAR ACCURACIES

2018: 73.4% • 2021: 86.0% • 2024: 86.5%

Per-Class Precision, Recall & F1-Score

Class Name	Precision	Recall	F1-Score	Support
Built-up	58.9%	59.3%	59.1%	123
Vegetation	84.1%	86.6%	85.3%	402
Water	100.0%	98.9%	99.5%	378
Agriculture	73.4%	78.3%	75.8%	240
Open land	47.3%	35.4%	40.5%	99

5 × 5 Confusion Matrix

True \ Pred	Built	Veg	Water	Agri	Open
Built-	73	8	0	19	23
Vegeta	6	348	0	35	13
Water	3	0	374	0	1
Agricu	14	36	0	188	2
Open l	28	22	0	14	35



2020–2024 Headline Urban Expansion

2020–2024 EXPANSION RANGE

+62.7 to +92.1 km²

Across Processing Methods

ESA WORLDCOVER 2021 ANCHOR

378.08 km²

Independent Benchmark (2021)

2024 CLEANED BUILT-UP FOOTPRINT

448.31 km²

Analysis Window: 2020–2024

Framing Note: Analysis is framed over **2020–2024**. Pre-2020 observations (2018–2019) are omitted due to cloud coverage and early baseline variability. The year **2022** is designated *Oct-Dec 2021, Collection 1* (dry-season slice from early Collection 1 archive; scale and offset are derived directly from STAC item metadata).

Multi-Series Growth & Method-Sensitivity Band (2020–2024)

Year	Cleaned Series (km ²)	Raw Classified (km ²)	TLS Normalised (km ²)
2020	356.19 km ²	332.13 km ²	407.12 km ²
2021	421.99 km ²	386.55 km ²	386.55 km ²
2022 (Oct-Dec 2021, Collection 1)	443.75 km ²	692.99 km ²	393.07 km ²
2023	479.78 km ²	359.65 km ²	440.15 km ²
2024	448.31 km ²	406.65 km ²	469.80 km ²

Leave-One-Year-Out (LOYO) Validation (Evaluated on ALL Held-Out Points)

To evaluate temporal stability without data leakage, classifiers were trained excluding each fold year, and tested on **all held-out points** (not restricted to stable points). Area estimates are adjusted using Olofsson et al. (2014) area-weighted stratified estimation with 95% confidence intervals.

Held-Out Year	Processing Stage	Precision	Recall	Built-up F1	Mapping Accuracy
2018	Raw (Before)	28.3%	36.6%	0.3191	0.3191
2018	TLS Normalized (After)	28.8%	36.6%	0.3226	0.3226
2021	Raw (Before)	62.5%	61.0%	0.6173	0.6173
2021	TLS Normalized (After)	68.4%	63.4%	0.6582	0.6582
2024	Raw (Before)	50.0%	63.4%	0.5591	0.5591
2024	TLS Normalized (After)	49.1%	70.7%	0.5800	0.5800

⚠ Negative Result Notice (Radiometric Normalisation):

Cross-year Total Least Squares (TLS) pseudo-invariant feature radiometric normalisation was implemented and systematically tested. Empirical evaluation across all held-out test points demonstrates that **radiometric normalisation did not reduce year-to-year drift** compared to the baseline composite series. Consequently, rule-based temporal consistency filtering remains the authoritative operational mechanism for ensuring monotonic urban expansion.

4-Stratum Change Validation (2020–2024)

Not independently validated. (Pune model evaluation is performed via held-out spatial block cross-validation against WorldCover).

Spatial Sprawl, Entropy & Core-Periphery Distribution

Year	Cleaned Built-up (km ²)	Shannon Entropy (H_n)	Core Share (0-6 km)
2020	332.13 km ²	0.9535	14.7
2021	386.55 km ²	0.9570	13.7
2022 (Oct-Dec 2021, Collection 1)	692.99 km ²	0.9562	13.7
2023	359.65 km ²	0.9567	12.0
2024	406.65 km ²	0.9565	13.0



5. Data Quality, Diagnostics & Quality Gate

Every composite is strictly validated for cloud contamination, NoData gaps, radiometric calibration shifts, and year-to-year area volatility before ingestion.

Ahmedabad: Satellite Diagnostics

Year	Scenes	Scene Dates	MGRS Tiles	NoData %	Cloud %	Stable Red
2018	8	2017-12-29, 2018-01-03, 2018-02-12, 2018-03-14	42QZL, 43QBF	0.00%	0.23%	0.1455
2019	8	2018-10-15, 2018-10-25, 2018-11-04, 2019-03-24, 2019-03-29	42QZL, 43QBF	0.00%	0.00%	0.1667
2020	8	2020-01-23, 2020-02-17, 2020-02-22, 2020-02-27, 2020-03-03, 2020-03-08	42QZL, 43QBF	0.00%	0.00%	0.1572
2021	8	2020-11-23, 2020-12-18, 2021-01-12, 2021-02-01, 2021-02-21, 2021-02-26	42QZL, 43QBF	0.00%	0.00%	0.1508
2022	8	2021-10-19, 2021-11-13, 2021-12-18, 2021-12-23	42QZL, 43QBF	0.00%	0.00%	0.1494
2023	8	2022-12-28, 2023-01-22, 2023-01-27, 2023-02-06, 2023-03-03, 2023-03-18	42QZL, 43QBF	0.00%	0.00%	0.1512
2024	8	2023-11-18, 2023-12-23, 2024-01-12, 2024-03-12, 2024-03-22	42QZL, 43QBF	0.00%	0.00%	0.1580

Data Quality Gate Status: **FAILED**

- **Gate 1 (Max NoData $\leq$ 5.0%):** **PASS** — All composites within nominal threshold ($\leq$ 5%)
- **Gate 2 (Max YoY Volatility $\leq$ 15.0%):** **FAIL** — Exceeded in: 2019 (25.7%)
- **Gate 3 (Min Accuracy $\geq$ 70.0%):** **FAIL** — Sub-70% in: 2018 (59.5%), 2021 (68.5%), 2024 (67.8%)
- **Gate 4 (Loss/Gain Ratio $\leq$ 30.0%):** **PASS** — Transition loss verified

Pune: Satellite Diagnostics

Year	Scenes	Scene Dates	MGRS Tiles	NoData %	Cloud %	Stable Red
2018	5	2018-03-06, 2018-03-31, 2018-05-25	43QCA, 43QDA	0.00%	8.28%	0.1452
2019	3	2019-01-05, 2019-01-25	43QCA, 43QDA	0.00%	2.59%	0.1163
2020	8	2019-12-31, 2020-01-30, 2020-02-14, 2020-02-19, 2020-02-24	43QCA, 43QDA	0.00%	0.00%	0.1175
2021	8	2020-11-10, 2020-12-05, 2020-12-25, 2021-01-24, 2021-02-03, 2021-02-28	43QCA, 43QDA	0.00%	0.00%	0.1147
2022	8	2021-11-10, 2021-12-10, 2021-12-20, 2021-12-25	43QCA, 43QDA	0.00%	2.81%	0.1127
2023	8	2022-12-20, 2022-12-25, 2023-01-14, 2023-01-19, 2023-02-03	43QCA, 43QDA	0.00%	0.00%	0.1100
2024	8	2023-11-15, 2024-02-03, 2024-02-18, 2024-02-23, 2024-02-28	43QCA, 43QDA	0.00%	0.00%	0.1225

Data Quality Gate Status: FAILED

- **Gate 1 (Max NoData $\leq$ 5.0%): PASS** — All composites within nominal threshold ($\leq$ 5%)
- **Gate 2 (Max YoY Volatility $\leq$ 15.0%): FAIL** — Exceeded in: 2019 (31.7%), 2021 (18.5%)
- **Gate 3 (Min Accuracy $\geq$ 70.0%): PASS** — All evaluated years $\geq$ 70.0%

- **Gate 4 (Loss/Gain Ratio $\leq$ 30.0%):** **PASS** — Transition loss verified



6. Engineering & Operations

Technology Stack

Layer	Technology	Purpose
EO Ingestion	PySTAC-Client, Stackstac, Dask	Cloud-native STAC scene retrieval & median reduction
ML Pipeline	Scikit-Learn, Rasterio, GeoPandas	Balanced Random Forest classification & spatial filtering
Spatial DB	PostgreSQL 16 + PostGIS 3.4	Spatial envelopes, audit runs, time-series storage
Backend API	FastAPI, Uvicorn, Pydantic	High-throughput REST API with GeoJSON endpoints
Frontend UI	MapLibre GL JS, Vanilla HTML/CSS	WebGL swipe map, time slider, interactive analytics
Orchestration	k3s / k3d, Docker Compose	Lightweight container & Kubernetes cluster parity
CI/CD	GitHub Actions	Automated linting, test execution, Trivy CVE scanning
Monitoring	Prometheus, Grafana	Custom .prom metrics exporter & sprawl monitoring

Test Suite & Payload Metrics

AUTOMATED UNIT & INTEGRATION TESTS

93 Test Cases

100% Passing Coverage across Scaling, Cleanup, Gate & Metrics

UrbanPulse is 100% configuration-driven. Add configs/<city>.yaml with the exact keys:

```
city:
  name: Jaipur
  state: Rajasthan
  country: India
spatial:
  bbox: [75.6000, 26.7000, 76.0500, 27.1500]
  crs: EPSG:4326
  center_lat: 26.9124
  center_lon: 75.7873
  aoi_size_km: 45.0
temporal:
  analysis_years:
    start_year: 2018
    end_year: 2024
  dry_season:
    start_month: 11
    end_month: 2
    start_day: 11-01
    end_day: 02-28
stac:
  earth_search_url: https://earth-search.aws.element84.com/v1
  collections:
    sentinel_2: sentinel-2-c1-l2a
```

7. Limitations

- accuracy is agreement with ESA WorldCover, not field-verified ground truth
- Sentinel-2 covers 2017 onward only
- dry-season composites confuse fallow farmland with built-up land
- the model is not transferable across sensors
- the persistence rule forces non-decreasing built-up area and is a documented assumption
- 10-60 m resolution limits small features.



8. Future Roadmap

1. Historical Landsat Integration (2000–2016)

Extend the temporal horizon by 16 years using harmonized Landsat 5/7/8 surface reflectance stacks with cross-calibration regression.

2. CA-Markov 2035 Urban Growth Forecasting

Couple Cellular Automata and Markov Transition Chains with slope, transport proximity, and zoning constraint layers for 2035 sprawl forecasting.

3. Hand-Labelled Ground-Truth Validation Set

Digitize high-resolution aerial imagery ground truth across 500 stratified random points to measure real-world precision beyond global maps.

4. Dynamic "Run This City" API Endpoint

Expose asynchronous Celery / Redis worker endpoints allowing users to submit any arbitrary bounding box for on-demand sprawl computation.

9. Reproducibility & Verification

All pipelines, tests, and documentation artifacts can be executed locally or in containerized environments with standard commands:

```
# 1. Start full local stack (Web + API + DB + Prometheus)
make up

# 2. Run complete pipeline for a specific city and target year
make pipeline CITY=pune YEAR=2024

# 3. Execute standalone Python pipeline CLI
python -m pipeline.run_city --city ahmedabad

# 4. Run automated test suite
pytest -v

# 5. Re-generate this documentation and PDF report
make docs
```

Component	Version / Status
Python	3.14.2
Rasterio	1.5.1
GeoPandas	1.1.4
Scikit-Learn	1.8.0
Numpy	2.4.4
Pandas	2.3.3
Dask	2026.8.0
Stackstac	0.5.1
PySTAC-Client	0.9.0
Matplotlib	3.10.9
Playwright	Available